PART II

LINEAR ESTIMATION

CHAPTER 3

Normal Equations

In Secs. 1.2 and 2.1 we studied the problem of determining the optimal function $h(\cdot)$ that minimizes the mean-square error of estimating a random variable $\boldsymbol{x}$ from another random variable $\boldsymbol{y}$. Specifically, we solved

$$\min_{h(\cdot)} \ \mathsf{E}\,\tilde{\boldsymbol{x}}\tilde{\boldsymbol{x}}^* \tag{3.1}$$

over all functions $h(\cdot)$ of $\boldsymbol{y}$. The optimal solution was found to be the conditional expectation of $\boldsymbol{x}$ given $\boldsymbol{y}$, i.e.,

$$\hat{\boldsymbol{x}} = \mathsf{E}\,(\boldsymbol{x}|\boldsymbol{y})$$

Such conditional expectations are generally hard to evaluate in closed-form, except in some special cases. We encountered three such cases in Part I (*Optimal Estimation*), namely the case of a BPSK signal embedded in additive Gaussian noise (Ex. 1.2), the case of jointly Gaussian random variables (studied in Secs. 1.4 and 2.2), and the case of random variables with exponential distributions (studied in Prob. I.16). Due to the difficulty in evaluating $\mathsf{E}\,(\boldsymbol{x}|\boldsymbol{y})$ in general, it is common practice to restrict the choice of $h(\cdot)$ to the subclass of *affine* functions of $\boldsymbol{y}$, i.e., to functions of the form

$$h(\boldsymbol{y}) \ = \ K\boldsymbol{y} + b \tag{3.2}$$

for some matrix K and for some vector b to be determined. For general vector-valued variables $\{\boldsymbol{x}, \boldsymbol{y}\}$, K is a matrix and b is a vector. When $\{\boldsymbol{x}, \boldsymbol{y}\}$ are scalars, then $\{K, b\}$ will also be scalars. If $\boldsymbol{x}$ is a scalar and $\boldsymbol{y}$ is a vector, then K will be a row vector and b a scalar. In some applications, the variables $\{\boldsymbol{x}, \boldsymbol{y}\}$ may be matrix-valued, in which case $\{K, b\}$ will be matrix-valued. In other words, the dimensions of $\{K, b\}$ need to be consistent with the dimensions of $\{\boldsymbol{x}, \boldsymbol{y}\}$, and these dimensions will be obvious from the context.

Affine functions of the form (3.2) are easier to implement than most nonlinear functions. For example, when K is a row vector, the product $K\boldsymbol{y}$ amounts to an inner product. Moreover, by deliberately restricting $h(\cdot)$ to be an affine function of $\boldsymbol{y}$, the evaluation of the optimal $h(\cdot)$ is greatly simplified. In particular, it will be seen that all we need to know in order to determine the optimal parameters $\{K, b\}$ are the first and second-order moments of $\{\boldsymbol{x}, \boldsymbol{y}\}$, namely, $\mathsf{E}\,\boldsymbol{x}, \mathsf{E}\,\boldsymbol{y}, \mathsf{E}\,\boldsymbol{x}\boldsymbol{x}^*, \mathsf{E}\,\boldsymbol{y}\boldsymbol{y}^*$, and $\mathsf{E}\,\boldsymbol{x}\boldsymbol{y}^*$. No other moments are needed. In contrast, evaluation of the optimal estimator, $\hat{\boldsymbol{x}} = \mathsf{E}\,(\boldsymbol{x}|\boldsymbol{y})$, requires full knowledge of the conditional pdf $f_{\boldsymbol{x}|\boldsymbol{y}}(x|y)$.

Of course, the minimum mean-square error (m.m.s.e.) that results from using the affine estimator (3.2) will be larger than the m.m.s.e. that results from the optimal design (3.1). One notable exception is the case of jointly Gaussian random variables $\{\boldsymbol{x}, \boldsymbol{y}\}$. It was verified in Sec. 2.2 that for Gaussian random variables, the optimal estimator is an affine function of the observation (so that, in this case, the affine and optimal estimators achieve the same m.m.s.e.) – see Lemma 2.1. Still, in general and for other signal distributions, the performance of the affine estimator (3.2) is reasonable enough for many important applications, and this fact explains its widespread adoption in current practice.

In this chapter we shall study the linear (affine) estimation problem in some detail. Before proceeding, we would like to remind the reader that the presentation in Part I (*Optimal Estimation*) has shown that all results for scalar real-valued random variables could be extended rather immediately to vector and also complex-valued random variables by relying on the vector and complex conjugation notation. For this reason, we shall deal directly with the vector and complex-valued case in the sequel.

3.1 MEAN-SQUARE ERROR CRITERION

In order to simplify the presentation, we assume first that $\{\boldsymbol{x}, \boldsymbol{y}\}$ are *zero-mean* random variables. Later in Sec. 4.4 we show how the nonzero-mean case can be accommodated through the process of centering the random variables.

Thus, consider two zero-mean vector-valued random variables $\boldsymbol{x}$ and $\boldsymbol{y}$ and let

$$\boxed{\bar{x} \triangleq \mathsf{E}\,\boldsymbol{x} = 0, \quad \bar{y} \triangleq \mathsf{E}\,\boldsymbol{y} = 0, \quad R_x \triangleq \mathsf{E}\,\boldsymbol{x}\boldsymbol{x}^*, \quad R_y \triangleq \mathsf{E}\,\boldsymbol{y}\boldsymbol{y}^*, \quad R_{xy} \triangleq \mathsf{E}\,\boldsymbol{x}\boldsymbol{y}^*}$$

The dimensions of $\boldsymbol{x}$ and $\boldsymbol{y}$ need not be identical, say, $\boldsymbol{x}$ is $p\times 1$ and $\boldsymbol{y}$ is $q \times 1$. The analysis can be adjusted to treat the case of matrix-valued data as well, say, where $\boldsymbol{x}$ is $p \times r$ and $\boldsymbol{y}$ is $q \times s$ with r and s larger than one. However, it is common (and also sufficient) to illustrate the main ideas by focusing on the case of column vectors $\boldsymbol{x}$ and $\boldsymbol{y}$. In this case, $\{R_x, R_y, R_{xy}\}$ are $\{p \times p, q \times q, p \times q\}$ matrices, respectively.

We now seek an affine estimator for $\boldsymbol{x}$, namely, one of the form

$$\hat{\boldsymbol{x}} = K\boldsymbol{y} + b$$

for some constants $\{K, b\}$ to be determined, where K is $p \times q$ and b is $p \times 1$. The determination of $\{K, b\}$ is based on two considerations. First, the estimator should be unbiased, which means that it should satisfy

$$\boxed{\mathsf{E}\,\hat{\boldsymbol{x}} = 0}$$

But since

$$\mathsf{E}\,\hat{\boldsymbol{x}} = K\mathsf{E}\,\boldsymbol{y} + b \;=\; 0 + b = b$$

we find that we must have $b = 0$. This means that the estimator that we are seeking is effectively a linear estimator, i.e., it is one of the form $\hat{\boldsymbol{x}} = K\boldsymbol{y}$. Second, the coefficient matrix K should be chosen optimally so as to minimize the error covariance matrix (or its trace), as we now explain.

Let $\{k_i^*\}$ denote the individual rows of K. Then the estimator for each entry of $\boldsymbol{x}$, say, $\boldsymbol{x}(i)$, is given by the inner product $k_i^*\boldsymbol{y}$,

$$\underbrace{\begin{bmatrix} \hat{\boldsymbol{x}}(0) \\ \hat{\boldsymbol{x}}(1) \\ \vdots \\ \hat{\boldsymbol{x}}(p-1) \end{bmatrix}}_{=\hat{\boldsymbol{x}}} = \begin{bmatrix} k_0^*\boldsymbol{y} \\ k_1^*\boldsymbol{y} \\ \vdots \\ k_{p-1}^*\boldsymbol{y} \end{bmatrix} = \underbrace{\begin{bmatrix} \rule{2em}{0.4pt} \\ \rule{2em}{0.4pt} \\ \vdots \\ \rule{2em}{0.4pt} \end{bmatrix}}_{=K} \boldsymbol{y}$$

The optimal choices for the column vectors $\{k_i\}$ are determined by solving

$$\boxed{\min_{k_i} \ \mathsf{E}\,|\tilde{\boldsymbol{x}}(i)|^2} \qquad \text{for each } i = 0, 1, \ldots, p-1 \tag{3.3}$$

where

$$\tilde{\boldsymbol{x}}(i) = \boldsymbol{x}(i) - \hat{\boldsymbol{x}}(i)$$

is the estimation error for the i-th entry of $\boldsymbol{x}$. That is, each column vector k_i is determined by minimizing the error variance of the corresponding entry in $\boldsymbol{x}$. The optimization problems (3.3) can be grouped together and stated equivalently as the problem of determining the matrix K by solving

$$\boxed{\min_{K} \ \mathsf{E}\,\tilde{\boldsymbol{x}}^*\tilde{\boldsymbol{x}}} \tag{3.4}$$

where $\tilde{\boldsymbol{x}} = \boldsymbol{x} - \hat{\boldsymbol{x}}$. This is because the scalar quantity $\mathsf{E}\,\tilde{\boldsymbol{x}}^*\tilde{\boldsymbol{x}}$ in (3.4) is simply the sum of the individual error variances that appear in (3.3),

$$\mathsf{E}\,\tilde{\boldsymbol{x}}^*\tilde{\boldsymbol{x}} \;=\; \mathsf{E}\,|\tilde{\boldsymbol{x}}(0)|^2 \;+\; \mathsf{E}\,|\tilde{\boldsymbol{x}}(1)|^2 \;+\; \ldots \;+\; \mathsf{E}\,|\tilde{\boldsymbol{x}}(p-1)|^2 \tag{3.5}$$

and each term $\mathsf{E}\,|\tilde{\boldsymbol{x}}(i)|^2$ depends on the corresponding k_i alone. In this way, minimizing $\mathsf{E}\,\tilde{\boldsymbol{x}}^*\tilde{\boldsymbol{x}}$ over K is equivalent to minimizing each term $\mathsf{E}\,|\tilde{\boldsymbol{x}}(i)|^2$ over its k_i, so that problems (3.3) and (3.4) are equivalent. We now proceed slowly and show how to solve (3.3) over the corresponding *vector* arguments $\{k_i\}$. Later in Sec. 3.4, we shall show how to minimize $\mathsf{E}\,\tilde{\boldsymbol{x}}^*\tilde{\boldsymbol{x}}$ directly over the matrix argument K, rather than in steps over the individual $\{k_i\}$.

Therefore, continuing with (3.3), we expand the cost function to obtain a *quadratic* expression in the unknown column vector k_i,

$$\begin{aligned}
\mathsf{E}\,|\tilde{\boldsymbol{x}}(i)|^2 \;&\triangleq\; \mathsf{E}\,|\boldsymbol{x}(i) - k_i^*\boldsymbol{y}|^2 \\
&=\; \mathsf{E}\,[\boldsymbol{x}(i) - k_i^*\boldsymbol{y}][\boldsymbol{x}(i) - k_i^*\boldsymbol{y}]^* \\
&=\; \mathsf{E}\,|\boldsymbol{x}(i)|^2 - [\mathsf{E}\,\boldsymbol{x}(i)\boldsymbol{y}^*]k_i - k_i^*[\mathsf{E}\,\boldsymbol{y}\boldsymbol{x}^*(i)] + k_i^* R_y k_i
\end{aligned}$$

This is a *scalar-valued* cost function of a possibly *complex-valued vector* quantity, k_i. We denote it by

$$J(k_i) \;\triangleq\; \mathsf{E}\,|\boldsymbol{x}(i)|^2 - [\mathsf{E}\,\boldsymbol{x}(i)\boldsymbol{y}^*]k_i - k_i^*[\mathsf{E}\,\boldsymbol{y}\boldsymbol{x}^*(i)] + k_i^* R_y k_i \tag{3.6}$$

The quantity $\mathsf{E}\,|\boldsymbol{x}(i)|^2$ that appears in the expression for $J(k_i)$ is the variance of $\boldsymbol{x}(i)$ and is therefore equal to the i-th diagonal entry of R_x. We denote it by

$$\sigma_{x,i}^2 = \mathsf{E}\,|\boldsymbol{x}(i)|^2$$

Likewise, the quantity $\mathsf{E}\,\boldsymbol{x}(i)\boldsymbol{y}^*$ is the i-th row of the cross-covariance matrix R_{xy}. We denote it by $R_{xy,i} = \mathsf{E}\,\boldsymbol{x}(i)\boldsymbol{y}^*$. In this way, we can rewrite $J(k_i)$ as

$$\boxed{J(k_i) \;\triangleq\; \sigma_{x,i}^2 - R_{xy,i}k_i - k_i^* R_{yx,i} + k_i^* R_y k_i} \tag{3.7}$$

where $\{\sigma_{x,i}^2, R_{xy,i}, R_y\}$ are known quantities and k_i is the unknown column vector; $\sigma_{x,i}^2$ is a scalar, $R_{xy,i}$ is a row vector, and R_y is a nonnegative-definite matrix. Moreover, $R_{yx,i} = R_{xy,i}^*$. Our objective is to minimize $J(k_i)$ over k_i.

We can proceed in at least two different ways. One way relies on standard differentiation techniques from calculus, while the second way relies on a completion-of-squares argument. While it may be easier for the reader to assimilate the differentiation argument due to familiarity with the concept of derivatives and function minimization, the second argument on completion of squares will serve as a powerful example of the convenience of the vector notation. Since this argument will be useful in other scenarios as well, we choose to include it here for the benefit of the reader, in addition to the differentiation argument. Still, both arguments require careful reasoning, as we explain below. Later, when similar derivations are needed in other chapters, we shall move more swiftly through the presentation.

3.2 MINIMIZATION BY DIFFERENTIATION

We start with the differentiation argument. As the reader may recall from a basic course on calculus, in order to determine the vector k_i that minimizes $J(k_i)$, we need to differentiate $J(k_i)$ with respect to each one of the entries of k_i, namely, $\{k_{ij}, j = 0, 1, \ldots, q-1\}$ and set the derivatives equal to zero. The main complication that arises here, relative to what the reader may be familiar with from a standard course on calculus, is that the entries $\{k_{ij}\}$ are *complex-valued*, and we therefore need to explain what is meant by differentiating $J(k_i)$ with respect to a complex variable. This is done in Chapter. C, where we show how to differentiate a function with respect to a complex scalar and even a complex vector. When the rules derived in that appendix are applied to the cost function $J(k_i)$ in (3.7) we find that the complex gradient vector of $J(k_i)$ with respect to k_i is given by

$$\nabla_{k_i} J(k_i) \;=\; -R_{xy,i} + k_i^* R_y$$

Observe that this result is consistent with what we would expect from the standard rules of differentiation for functions of real variables, with the vectors k_i and k_i^* treated as different quantities. Also, if all data were real-valued, in which case

$$J(k_i) = \mathsf{E}\,\boldsymbol{x}^2(i) - [\mathsf{E}\,\boldsymbol{x}(i)\boldsymbol{y}^{\mathsf{T}}]k_i - k_i^{\mathsf{T}}[\mathsf{E}\,\boldsymbol{y}\boldsymbol{x}(i)] + k_i^{\mathsf{T}} R_y k_i$$

then we would have obtained instead $\nabla_{k_i} J(k_i) = -R_{xy,i} + 2k_i^{\mathsf{T}} R_y$, with an additional factor of 2 — see Chapter C.

By setting the complex gradient equal to zero at the optimal choice $k_i = k_i^o$, we find that k_i^o should satisfy the linear equations

$$k_i^{o*} R_y = R_{xy,i}, \qquad i = 0, 1, \ldots, p-1 \tag{3.8}$$

The vector k_i^o so obtained minimizes $J(k_i)$ since the Hessian matrix of $J(k_i)$ with respect to k_i is equal to R_y, which is a nonnegative-definite matrix. Hessian matrices are defined in Chapter C; they are obtained by further differentiating $\nabla_{k_i} J(k_i)$ with respect to k_i^*.

If we collect the row vectors $\{k_i^{o*}\}$ from (3.8) into a matrix K_o we find that this desired solution matrix should satisfy

$$\boxed{K_o R_y = R_{xy}} \tag{3.9}$$

These equations are called the normal equations, for reasons explained later in Remark 4.1 in the next chapter.

3.3 MINIMIZATION BY COMPLETION OF SQUARES

We now re-establish (3.9), from first principles, by using a completion-of-squares argument that avoids the need for dealing with complex gradients. Thus, consider the cost function

(3.7) and note that it can be expressed in matrix form as follows:

$$\boxed{J(k_i) = \begin{bmatrix} 1 & k_i^* \end{bmatrix} \begin{bmatrix} \sigma_{x,i}^2 & -R_{xy,i} \\ -R_{yx,i} & R_y \end{bmatrix} \begin{bmatrix} 1 \\ k_i \end{bmatrix}} \tag{3.10}$$

with a Hermitian center matrix and with the unknown vector k_i, and its conjugate transpose, multiplying from both sides.

Now given any Hermitian matrix of the form

$$M = \begin{bmatrix} A & B \\ B^* & C \end{bmatrix}$$

with $A = A^*$, $C = C^*$, and C invertible, it can be verified by direct calculation that M can be factored into a product of a block upper-triangular, block-diagonal, and block lower-triangular matrices as:

$$\boxed{\begin{bmatrix} A & B \\ B^* & C \end{bmatrix} = \begin{bmatrix} \mathrm{I} & BC^{-1} \\ 0 & \mathrm{I} \end{bmatrix} \begin{bmatrix} \Sigma & 0 \\ 0 & C \end{bmatrix} \begin{bmatrix} \mathrm{I} & 0 \\ C^{-1}B^* & \mathrm{I} \end{bmatrix}} \tag{3.11}$$

where

$$\Sigma \triangleq A - BC^{-1}B^*$$

is called the Schur complement of M with respect to C. This is a general matrix result; it is an immediate extension to Hermitian matrices of a similar factorization result that was described in a footnote in Sec. 1.4 for symmetric matrices. The validity of (3.11) can be verified by expanding the right-hand side. The factorization (3.11) for M is valid as long as the matrix C is invertible, which means that we cannot apply the result directly to the center matrix

$$\begin{bmatrix} \sigma_{x,i}^2 & -R_{xy,i} \\ -R_{yx,i} & R_y \end{bmatrix}$$

that appears in our expression (3.10) for $J(k_i)$. The reason being that the covariance matrix R_y is only required to be nonnegative-definite (and, hence, possibly singular). However, there is a generalization of (3.11) for block matrices M with possibly singular matrices C. Indeed, it is easy to verify, also by direct calculation, that we can alternatively factor any such M as

$$\boxed{\begin{bmatrix} A & B \\ B^* & C \end{bmatrix} = \begin{bmatrix} \mathrm{I} & D \\ 0 & \mathrm{I} \end{bmatrix} \begin{bmatrix} \Sigma & 0 \\ 0 & C \end{bmatrix} \begin{bmatrix} \mathrm{I} & 0 \\ D^* & \mathrm{I} \end{bmatrix}} \tag{3.12}$$

where D is defined as *any* solution to the linear system of equations

$$DC = B \tag{3.13}$$

and

$$\Sigma = A - BD^*$$

Clearly, when C is invertible, the above factorization reduces to (3.11). However, when C is singular, many solutions D may exist for (3.13) and, consequently, many factorizations of the form (3.12) may be possible for M. Our arguments will show that for the matrix M in question, namely, the center matrix in (3.10), a factorization of the form (3.12) always exists since the corresponding equations (3.13) will always have a solution D — see Sec. 4.3.

Applying (3.12) to the center matrix in (3.10) we can write

$$\begin{bmatrix} \sigma_{x,i}^2 & -R_{xy,i} \\ -R_{yx,i} & R_y \end{bmatrix} = \begin{bmatrix} 1 & -k_i^{o*} \\ 0 & \mathrm{I} \end{bmatrix} \begin{bmatrix} \sigma_{x,i}^2 - R_{xy,i}k_i^o & 0 \\ 0 & R_y \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -k_i^o & \mathrm{I} \end{bmatrix} \tag{3.14}$$

where k_i^o is any solution to the linear system of equations

$$k_i^{o*} R_y = R_{xy,i} \tag{3.15}$$

Substituting the factorization (3.14) into expression (3.10) for $J(k_i)$ and expanding the right-hand side, we find that $J(k_i)$ can be expressed in the equivalent form

$$\boxed{J(k_i) = (\sigma_{x,i}^2 - R_{xy,i}k_i^o) \; + \; (k_i - k_i^o)^* R_y (k_i - k_i^o)} \tag{3.16}$$

This is a revealing form for $J(k_i)$ since only the second term depends on the unknown k_i. But since R_y is nonnegative-definite, this second term is always nonnegative and it will be minimized (actually made equal to zero) by choosing $k_i = k_i^o$ (see Prob. II.5). We conclude that the minimizing k_i is given by any solution to (3.15). In addition, the resulting m.m.s.e. is given by any of the following equivalent forms:

$$\boxed{J(k_i^o) = \sigma_{x,i}^2 - R_{xy,i}k_i^o \; = \; \sigma_{x,i}^2 - k_i^{o*} R_y k_i^o \; = \; \sigma_{x,i}^2 - k_i^{o*} R_{yx,i}} \tag{3.17}$$

If we again collect the row vectors $\{k_i^{o*}\}$ into a matrix K_o we find that K_o should satisfy (3.9), namely, $K_o R_y = R_{xy}$. Moreover, the minimum value of the related cost (3.5) can be seen to be (see Probs. II.5 and II.6):

$$\boxed{\mathsf{E}\tilde{\boldsymbol{x}}^*\tilde{\boldsymbol{x}} \; = \; \mathsf{Tr}\,(R_x - K_o R_y K_o^*)} \tag{3.18}$$

3.4 MINIMIZATION OF THE ERROR COVARIANCE MATRIX

The same completion-of-squares argument can be applied directly to the solution of (3.4) rather than the solution of the individual problems (3.3). To see this, let $J(K)$ denote the error-covariance matrix,

$$\boxed{J(K) \; \triangleq \; \mathsf{E}\tilde{\boldsymbol{x}}\tilde{\boldsymbol{x}}^* \; = \; \mathsf{E}\,(\boldsymbol{x} - K\boldsymbol{y})(\boldsymbol{x} - K\boldsymbol{y})^*}$$

so that, as in (3.10),

$$J(K) = \begin{bmatrix} \mathrm{I} & K \end{bmatrix} \begin{bmatrix} R_x & -R_{xy} \\ -R_{yx} & R_y \end{bmatrix} \begin{bmatrix} \mathrm{I} \\ K^* \end{bmatrix}$$

Following the same arguments as in the previous section, we can factor the center matrix as

$$\begin{bmatrix} R_x & -R_{xy} \\ -R_{yx} & R_y \end{bmatrix} = \begin{bmatrix} \mathrm{I} & -K_o \\ 0 & \mathrm{I} \end{bmatrix} \begin{bmatrix} R_x - R_{xy}K_o^* & 0 \\ 0 & R_y \end{bmatrix} \begin{bmatrix} \mathrm{I} & 0 \\ -K_o^* & \mathrm{I} \end{bmatrix}$$

where K_o is any solution to $K_o R_y = R_{xy}$. It then follows that

$$J(K) \; = \; (R_x - R_{xy}K_o^*) \; + \; (K - K_o)R_y(K - K_o)^* \tag{3.19}$$

where the last term is nonnegative definite for any K since $R_y \geq 0$. Now the criterion in (3.4) is related to $J(K)$ via $\mathsf{E}\,\tilde{x}^*\tilde{x} = \mathsf{Tr}[J(K)]$ so that, from (3.19),

$$\mathsf{E}\,\tilde{x}^*\tilde{x} \geq \mathsf{Tr}(R_x - R_{xy}K_o^*) \qquad \text{for any } K \tag{3.20}$$

This is because the trace of the nonnegative-definite matrix $(K - K_o)R_y(K - K_o)^*$ is nonnegative. This result follows from the fact that the trace of a matrix, which is the sum of its diagonal elements, is also equal to the sum of its eigenvalues — see Prob. II.3. Now since the eigenvalues of a nonnegative matrix are necessarily nonnegative, it follows that $\mathsf{Tr}[(K - K_o)R_y(K - K_o)^*] \geq 0$. Equality in (3.20) is achieved by setting $K = K_o$ in (3.19).

This derivation reveals another aspect of the solution K_o. It not only minimizes the trace of the error covariance matrix, as in (3.4), but it also minimizes the error-covariance matrix itself since we also get $J(K) \geq J(K_o)$ for any K. We can therefore interpret K_o as also the solution to the problem

$$\boxed{\min_{K} \quad \mathsf{E}\,\tilde{x}\tilde{x}^*} \tag{3.21}$$

which is in terms of the error-covariance matrix, rather than its trace. The resulting minimum value is

$$\boxed{J(K_o) = R_x - R_{xy}K_o^*} \tag{3.22}$$

The optimization problem (3.21) is interesting for two reasons. First, the cost function $J(K) = \mathsf{E}\,\tilde{x}\tilde{x}^*$ is matrix-valued. That is, it assumes *matrix values* for each choice of K. Second, the unknown argument, K, is a *matrix* itself. In this way, problem (3.21) involves minimizing a matrix-valued cost function over a matrix-valued argument.

3.5 OPTIMAL LINEAR ESTIMATOR

We shall have more to say about the solution(s) K_o of the normal equations $K_oR_y = R_{xy}$. For now, we summarize the main conclusions of the last two sections.

Theorem 3.1 (Optimal linear estimator) Given zero-mean random variables x and y, the linear least-mean-squares estimator (l.l.m.s.e.) of x given y is

$$\hat{x} = K_o y$$

where K_o is any solution to the linear system of equations $K_oR_y = R_{xy}$. This estimator minimizes the following two error measures:

$$\min_K \mathsf{E}\,\tilde{x}^*\tilde{x} \quad \text{and} \quad \min_K \mathsf{E}\,\tilde{x}\tilde{x}^*$$

The scalar cost on the left is the trace of the matrix cost on the right. The resulting minimum mean-square errors, as defined by (3.4) and (3.22), are given by

$$\begin{aligned}
\min_K \mathsf{E}\,\tilde{x}^*\tilde{x} &= \mathsf{Tr}\,(R_x - K_oR_yK_o^*) \\
\min_K \mathsf{E}\,\tilde{x}\tilde{x}^* &= R_x - K_oR_yK_o^*
\end{aligned}$$